# PYRAMID QUESTIONS

1) Draw the following pyramids of numbers to scale on graph paper.

i) 15 hydrangea plants → 12 caterpillars → 1 willy wagtail
ii) 5 rose bushes → 100 greenfly → 25 ladybirds
iii) 50 lettuces → 20 slugs → 1 mudlark

a) Which one looks odd? Why does it have this shape?
b) Use the following information to draw a pyramid of biomass for each food chain

| Organism | Weight (each) | Organism | Weight (each) | Organism | Weight (each) |
|---|---|---|---|---|---|
| Hydrangea | 10g | Caterpillar | 1g | Willy wagtail | 10g |
| Rose bush | 250g | Greenfly | 0.5g | Ladybird | 1g |
| Lettuce | 10g | Slug | 1g | Mudlark | 10g |

c) Do any of them look odd? Why not?

2) The following information was collected by a class of pupils studying the numbers of animals feeding on a eucalypt tree.

| Animal | Number | Food which each animal eats |
|---|---|---|
| Moth larvae | 93 | Eucalypt leaves |
| Willy wagtail | 6 | Moth larvae |
| Possum | 3 | Eucalypt nuts |
| Hawk | 2 | pigeons and doves |
| Dove | 3 | Moth larvae |
| Crusader beetle | 5 | Moth larvae |

Copy the following table and use the information above to complete it.

| Trophic level | Names of organisms | Number of organisms | Energy / day |
|---|---|---|---|
| Tertiary consumers | | | 2 Kj |
| Secondary consumers | | | 18 Kj |
| Primary consumers | | | 165 Kj |
| Producers | | | 1770 Kj |

a) Draw a pyramid of energy - to scale - for the above results.
b) Why might the information collected by the children not give a completely accurate picture of the number of organisms in the eucalypt tree?